

Pore-scale modelling of complex conductivity of saturated granular materials

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ABSTRACT

The complex conductivity method has been frequently used in solving hydrogeological, engineering, and environmental problems in practice. Macroscopic geophysical responses are governed by pore-scale rock properties; therefore, a clear petrophysical understanding of how the pore-scale structure controls electric conduction and polarization in geomaterials is necessary to fully interpret laboratory-/field-scale geophysical observations. In this study, we have developed a pore-scale numerical approach to calculate the complex conductivity of water-saturated granular materials. This physics-based model has advantage over phenomenological or empirical models such that the intrinsic relations between pore structure and geoelectrical signals could be revealed. In the modelling, the influence of electrical double layer, which is quantified by complex surface conductance, can be converted to an apparent volumetric complex conductivity of either solid particles or pore fluid. The effective complex conductivity of the sample is determined by directly solving the finite-difference representation of the Laplace equation in the domain of a representative elementary volume. The numerical approach is tested on one synthetic sample and one glass beads sample. For the synthetic sample, the consistency between the numerical and the analytical solution confirms the accuracy of the approach. Comparison with experimental measurements of the glass beads sample reported previously indicates that the developed numerical approach can well reproduce the key characteristics of the complex conductivity of water-saturated granular materials in the frequency range between 10^{-1} and 10^4 Hz. Furthermore, the numerical examples also show that the proposed approach can capture the salinity-dependent electric conduction and polarization in saturated granular materials.

INTRODUCTION

Induced polarization (IP) is a useful geophysical technique for numerous environmental and hydrogeological applications (Kemna *et al.* 2012; Revil *et al.* 2012). In practice, IP measurement can be conducted in either the time domain or the frequency domain. When in the frequency domain, it is usually termed as spectral induced polarization (SIP), which measures the complex conductivity of a material. The real part of complex conductivity quantifies the ease of an electric current passing through the material, whereas the imaginary part of complex conductivity is a measure of the electric energy stored from the external electric field. The complex conductivity of a saturated geological material is controlled by electrical properties of individual phases, i.e., pore fluid and solid particles. The geometric characteristics of individual phases such as porosity, surface area, and particle/pore size also affect the effective complex conductivity of geomaterials (Börner and Schön 1991; Scott and Barker 2003; Revil, Florsch and Camerlynck 2014a).

In addition to solid particles and pore fluid, the solid–liquid interface could also influence the complex conductivity of geological materials at low-to-intermediate frequencies (from mHz to MHz). When a two-phase geomaterial is subjected to an external electric field, charges may accumulate at the solid–fluid interface because of electrical heterogeneity, giving rise to an overall separation between the negative and positive charges. Macroscopically, the material exhibits an electric polarization, termed as interfacial polarization (Hanai 1960) or Maxwell–Wagner polarization. This polarization, which occurs at intermediate frequencies (from kHz to MHz), is usually modelled with the differential effective medium (DEM) theory (e.g., Chen and Or 2006).

At low frequencies (<kHz), the electrical double layer (EDL) forming at the solid–liquid interface could significantly affect both the electric conduction and polarization. In the EDL, the ion density is usually larger than that in the bulk solution (Glover *et al.* 1994), and therefore, the electrical conduction along the liquid–solid interface could be enhanced (Revil and Glover 1998). In addition to electric conduction, the EDL also plays an impor-

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tant role in electric polarization (Revil 2012; Bucker and Hördt 2013). Two prevailing theories were developed to explain the role of EDL in controlling the IP response of geomaterials at low frequencies: EDL polarization (Schwarz 1962) and membrane polarization (Marshall and Madden 1959). Recent developments of these theories have incorporated electrochemical properties of the EDL. To a certain extent, both theories could explain the effect of fluid salinity, pH, temperature, and/or surface charge density on the imaginary conductivity of geological materials (Revil 2012, 2013; Bairlein *et al.* 2016; Hördt *et al.* 2016; Leroy *et al.* 2017). However, it is still unclear how these two models can be reconciled. Therefore, further studies in modelling IP responses of geomaterials are strongly needed.

Recent developments and advances in pore-scale imaging techniques such as X-ray tomography, scanning electron microscopy (SEM), and focused ion beam SEM have enabled the morphological visualisation of the microstructure of geomaterials (Fredrich 1999; Wildenschild and Sheppard 2013). Based on the digital representation of the microstructure, it is possible to compute a geomaterial's physical and transport properties from first principles. Related examples include but not limited to simulation of elastic moduli, electrical resistivity, dielectric constant, water retention curve, and permeability of various geological materials (Saenger *et al.* 2011; Madonna, Almqvist and Saenger 2012; Andrä *et al.* 2013a, b; Blunt *et al.* 2013). Our understanding of the physical and transport properties of porous media has been greatly advanced in recent years with the aid of pore-scale numerical modelling techniques.

However, the pore-scale modelling based on a digital representation of actual microstructures has not yet been applied to study the IP response of geological materials at low-to-intermediate frequencies. It is anticipated that the IP related pore-scale modelling can be used, for instance, to study the controversial controlling pore geometry parameters for the imaginary conductivity (Revil 2014; Weller, Slater and Nordsiek 2014; Niu *et al.* 2016a). It can also be used to test the existing complex conductivity models of geomaterials, such as Archie's model and its modified forms (Glover, Hole and Pous 2000), DEM theory-based models (Bussian 1983), and volume average-based models (Pride 1994; Revil 2013). Another potential application of pore-scale IP modelling is the joint calculation of IP and permeability. The simulated results, along with the digital representation of the material's microstructure, can be used to evaluate some IP-based permeability prediction models (Slater *et al.* 2014; Revil *et al.* 2015).

The objective of this study is to develop a pore-scale modelling technique to simulate the IP response (in terms of complex conductivity) of saturated geomaterials. For simplicity, only granular materials with spherical particles are considered here. We first briefly review the theories describing the influence of EDL on the electric conduction and polarization of geomaterials. The contribution of EDL is quantified by a surface property (complex surface conductance), which is then converted to an

apparent volumetric property for solid particle or pore fluid. Then, the effective complex conductivity of the granular material is numerically determined by solving the Laplace equation. Next, the numerical approach is tested on a synthetic sample with a simple cubic structure and on a glass beads sample reported in the work of Leroy *et al.* (2008). In the last section of this paper, we use this approach to analyse the salinity effect on the complex conductivity of water-saturated granular materials.

INFLUENCE OF THE ELECTRICAL DOUBLE LAYER

In soils and rocks, the mineral surface is usually negatively charged because of isomorphous substitutions (McBride 1989) and/or the presence of hydroxyl group (Zhuravlev 2000). When saturated with an electrolyte, ions will accumulate at the solid-liquid interface to balance the negatively charged mineral surface, forming the so-called EDL. In the inner layer of the EDL (i.e., Stern layer), the cations are adsorbed on the mineral surface. The outer layer (i.e., diffuse layer) extends from the outer plane of the Stern layer into the pore space. The electrical properties of the EDL are distinct from the bulk solution, and the EDL can significantly influence the effective electrical properties of the material.

Influence of the diffuse layer

In the EDL, ion density is larger than that in the bulk solution, and thus, electrical conduction can be enhanced at the solid-liquid interface. As discussed in the work of Revil and Glover (1998), both the diffuse layer and Stern layer could contribute to the surface conduction. In addition, the migration of protons on the mineral surface might also contribute to the surface conduction (O'Konski 1960; Skold, Revil and Vaudelet 2011; Leroy *et al.* 2013). In this study, we adopted the assumption used by Revil (2013). That is, the surface conductance at the low-frequency limit C_s^0 is solely from the diffuse layer expressed as

$$C_s^0 = \beta_{(+)} Q_d, \quad (1)$$

where Q_d and $\beta_{(+)}$ are the surface charge density and the ion mobility related to the diffuse layer. Note that equation (1) can be easily replaced by other models [e.g., equation (1) in the work of Revil and Glover (1998)] to consider other possible contributions.

The diffuse layer could also contribute to the electric polarization of a porous medium. Currently, there are two different mechanisms quantifying the role of the diffuse layer in the polarization. The first one addresses the influence of normal flow of ions in the diffuse layer on the Stern layer polarization (e.g., Lyklema, Dukhin and Shilov 1983); the second mechanism is related to the variation in ionic concentration in the diffuse layer induced by an external electric field (e.g., Shilov *et al.* 2001). In this study, only the first mechanism is considered and the details will be discussed in the next subsection along with the Stern layer polarization.

Influence of the Stern layer

The low-frequency IP response of geomaterials was usually interpreted by the theory of Stern layer polarization (e.g., Revil 2012) or membrane polarization (e.g., Bairlein *et al.* 2016). For granular materials characterised by large grain sizes, Stern layer polarization is most likely the controlling mechanism. In this section, we briefly review the key aspects of the Stern layer polarization model. Consider a spherical particle with radius r immersed in NaCl solution. After applying an electric field, the ions in the Stern layer will redistribute due to two driving forces: (1) electric field and (2) ion concentration gradient. The induced complex conductance C_s^s follows a simple Debye relaxation expressed as (Schwarz 1962)

$$C_s^s = \frac{i\omega\tau}{1+i\omega\tau} \Sigma_s^s, \quad (2)$$

where Σ_s^s is the surface conductance associated with the Stern layer, $\omega = 2\pi f$ is the angular frequency (with f being the frequency), and $i^2 = -1$. Parameter Σ_s^s has the following form:

$$\Sigma_s^s = \beta_{(+)}^s Q_s, \quad (3)$$

where Q_s and $\beta_{(+)}^s$ are the surface charge density and the ion mobility related to the Stern layer. The relaxation time is related to the radius of the curvature of the EDL r_c by (Schwarz 1962; Niu and Zhang 2017)

$$\tau = \frac{r_c^2}{2D^+}, \quad (4)$$

where D^+ is the diffusion coefficient of the ions in the Stern layer. For granular materials considered in this study, the EDL is coating on the particle surface, and therefore, r_c is equal to the particle radius r_d .

Equation (4) does not consider the polarization of the diffuse layer. To account for the effect, we adopted the diffuse layer polarization model developed by Lyklema *et al.* (1983), and thus, equation (4) can be adapted as (Niu and Revil 2016)

$$\tau = \frac{r_d^2}{2MD^+}. \quad (5)$$

Coefficient M can be related to the electrochemical properties of the EDL by

$$M = 1 - \frac{2}{\cosh\left(\frac{e\phi_d^0}{2k_bT}\right)} \left[\frac{Q_0\kappa}{4eC_f} - \sinh\left(\frac{e\phi_d^0}{2k_bT}\right) \right], \quad (6)$$

where $Q_0 = -Q_s - Q_d$ is the total surface charge density, e denotes the elementary charge, T is the absolute temperature, k_b is Boltzmann's constant, and C_f is the electrolyte concentration. The potential at the boundary between the Stern and diffuse layers ϕ_d^0 can be related to Q_d by the Gouy–Chapman relationship as follows:

$$Q_d = -\sqrt{8\varepsilon_w\varepsilon_0k_bTC_f} \sinh\left(\frac{e\phi_d^0}{2k_bT}\right), \quad (7)$$

where ε_w is the relative permittivity of the electrolyte and ε_0 is the vacuum permittivity. The Debye length κ^{-1} can be determined as

$$\kappa^{-1} = \sqrt{\frac{\varepsilon_w\varepsilon_0k_bT}{2e^2C_f}}. \quad (8)$$

Upscaling complex surface conductance

By combining the contributions from the diffuse layer and Stern layer, the EDL-induced complex surface conductance C_s^* can be then written as

$$C_s^* = C_s^0 + C_s^s. \quad (9)$$

Note that C_s^* is a surface property of the EDL with the unit in Siemens. Assume that the particle with the EDL is electrically equivalent to a particle without the EDL but with apparent complex volumetric conductivity σ_s^* (in S/m). The surface conductance can be then upscaled as

$$\sigma_s^* = \frac{2C_s^*}{\Pi}, \quad (10)$$

where Π is a characteristic particle size (Niu *et al.* 2016a). For spherical particles, Π is equal to the particle radius r_d (O'Konski 1960; Schwarz 1962). By using equation (10), the surface conductance (in S) can be then upscaled as volumetric conductivity (in S/m) for the solid phase (grains). The term “grain upscaling” is used in this study to represent the above upscaling process, and the word “grain” addresses the solid phase that gains apparent volumetric conductivity σ_s^* .

The complex surface conductance can also be treated as a perturbation of the fluid conductivity, and the increased complex conductivity $\Delta\sigma_w^*$ is then expressed as (Johnson, Koplik and Schwartz 1986; Niu *et al.* 2016a)

$$\Delta\sigma_w^* = \frac{2C_s^*}{\Lambda}, \quad (11)$$

where Λ is a characteristic pore size (Johnson *et al.* 1986), and it can be related to surface-area-to-pore-volume ratio S_{por} and the formation factor F by

$$\frac{2}{\Lambda} = -\frac{d\ln F}{d\ln\phi} S_{por}, \quad (12)$$

where ϕ is the porosity. If using Archie's equation $F = \phi^m$ (Archie 1942), equation (12) can be simplified as

$$\frac{2}{\Lambda} = mS_{por}, \quad (13)$$

where m is the cementation factor. Using equation (11), the EDL-induced complex surface conductance (in S) can be then converted to an apparent volumetric (complex) conductivity (in S m⁻¹) for pore fluid. Therefore, the complex conductivity of the fluid phase after upscaling becomes $\sigma_w + \Delta\sigma_w^*$. Similarly, we use the term “pore upscaling” to represent the above upscaling process, and the

word “pore” addresses the fluid phase that gains apparent volumetric conductivity $\Delta\sigma_w^*$. Note that, in “pore upscaling”, only the fluid phase gains additional conductivity while the conductivity of the solid phase is not affected by the EDL.

EFFECTIVE COMPLEX CONDUCTIVITY CALCULATION

In the last section, the EDL-induced complex surface conductance has been converted to an apparent conductivity of the solid particle or a perturbation in the fluid conductivity. Therefore, the geomaterial with solid, liquid, and EDL can be then simplified as a mixture of only solid particles and fluid. In this section, we discuss how to calculate the effective complex conductivity of this two-phase mixture.

Definition of effective complex conductivity in heterogeneous material

Consider the laboratory measurement of the electrical conductivity of an isotropic material. The applied current I flows through a sample, and the resulting potential difference ΔP across the sample is measured. Considering the sample's geometry, the average electrical field $\langle \mathbf{E} \rangle$ and current density $\langle \mathbf{J} \rangle$ can be calculated, and then the effective conductivity σ_{eff}^* of the material is determined from

$$\langle \mathbf{J} \rangle = \sigma_{eff}^* \langle \mathbf{E} \rangle. \quad (14)$$

Note that $\langle \mathbf{J} \rangle$ and $\langle \mathbf{E} \rangle$ usually are not in phase because of the storage of electric energy in the material, and therefore, the effective conductivity σ_{eff}^* is a complex number. Its in-phase and quadrature components are σ_{eff}' and σ_{eff}'' , respectively. In a heterogeneous material, the electric field $\mathbf{E}$ and the resulting current density $\mathbf{J}$ are not uniform, and they are functions of the spatial position $\mathbf{x} = (x, y, z)$ where x , y , and z are coordinates. Their relations to $\langle \mathbf{E} \rangle$ and $\langle \mathbf{J} \rangle$ are as follows (Mendelson 1975; Cheng and Torquato 1997):

$$\langle \mathbf{E} \rangle = -\frac{1}{V} \int_V \nabla \Psi(\mathbf{x}) dV \quad (15)$$

$$\langle \mathbf{J} \rangle = \frac{1}{V} \int_V \mathbf{J}(\mathbf{x}) dV, \quad (16)$$

where V is the domain of a representative elementary volume of the sample. The local current density $\mathbf{J}(\mathbf{x})$ and potential $\Psi(\mathbf{x})$ in the heterogeneous material can be determined by solving the Laplace equation as follows:

$$\nabla \mathbf{J}(\mathbf{x}) = -\nabla [\sigma^*(\mathbf{x}) \nabla \Psi(\mathbf{x})] = 0, \quad (17)$$

where $\sigma^*(\mathbf{x})$ is the local complex conductivity of the sample. If the location $\mathbf{x}$ is within the pore space, $\sigma^*(\mathbf{x})$ will have the same conductivity value with the pore fluid; otherwise, it has the electrical property of the solid phase.

Numerical calculation of σ_{eff}^*

To calculate the effective conductivity σ_{eff}^* of a heterogeneous material, the first step is to determine the local complex conductivity of the sample. In “grain upscaling”, the fluid conductivity is $\sigma_w + i\epsilon_w \epsilon_0 \omega$ where ϵ_w is the relative permittivity of the fluid (the term $i\epsilon_w \epsilon_0 \omega$ denotes the contribution of dielectric effect), and the apparent conductivity of the solid phase is the sum of the surface conductivity σ_s^* [equation (10)] and the dielectric effect, i.e., $i\epsilon_s \epsilon_0 \omega$, where ϵ_s is the relative permittivity of solid particles. In pore upscaling, the conductivity of the solid phase is only arisen from the dielectric effect with a value of $i\epsilon_s \epsilon_0 \omega$, whereas the fluid has an apparent conductivity of $\sigma_w + i\epsilon_w \epsilon_0 \omega + \Delta\sigma_w^*$, the last term of which is calculated with equation (11).

At a given frequency, once the local complex conductivity is defined, a potential gradient can be applied across the sample. The spatial distribution of electric potential Ψ and current density $\mathbf{J}$ in the sample can be then determined by solving equation (17) with the applied boundary conditions. Afterwards, the volume-averaging electric potential and current density, i.e., $\langle \mathbf{E} \rangle$ and $\langle \mathbf{J} \rangle$, can be determined according to equations (15) and (16). Finally, the effective conductivity of the sample σ_{eff}^* can be calculated with equation (14). It is noted that, since the local conductivity varies with frequency [e.g., see equations (2) and (10)], the effective conductivity of the sample is also frequency dependent.

The analytical solution to equation (17) only exists for heterogeneous materials with very simple microstructures. More generally, the equation has to be solved numerically to obtain the effective complex conductivity of the sample. In this study, we adopted an industrial standard code (AC3D.F) developed in the National Institute of Standards and Technology (available at <http://ciks.cbt.nist.gov/garboocz/manual/node52.html>). The code has been widely used to determine the effective conductivity/permittivity of heterogeneous materials (e.g., Torrents, Mason and Garboczi 2000; Kidner *et al.* 2002). It solves the finite-difference (FD) representation of the Laplace equation with a conjugate gradient method. Periodic boundary conditions are imposed during the calculation (e.g., see Guibert *et al.* 2016). Both isotropic and anisotropic materials can be handled by the code. For details of the algorithm, one can refer to Garboczi (1998). In the calculation, the sample is discretised into a number of cells, and each cell has a uniform electrical property. We modified the code to allow for dynamic allocation of memory, and thus, the model resolution can be increased with a given sample size. We also changed the format of the input files such that the complex conductivity at each cell can be assigned individually for the calculation.

Validation of the numerical calculation

To validate the numerical calculation, a synthetic sample with the simple cubic structure is considered (Figure 1). In each unit element of the lattice structure, there is a sphere in the centre of the cube. Assume that the background conductivity of the sample (i.e., fluid phase) is σ_1^* and the conductivity of the sphere (i.e.,

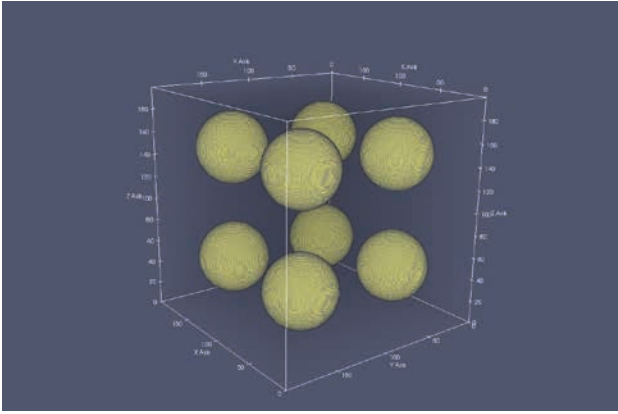


Figure 1 The 3D illustration of the simple cubic structure.

solid phase) is σ_2^* . Numerical simulations were performed to calculate the effective conductivity of the sample in Figure 1. A unit element is considered, and the particle radius is $35 \mu\text{m}$. In the calculation, we assume that the imaginary conductivity of each phase is only from dielectric polarization. Then, the complex conductivity σ^* of each phase (e.g., solid phase or liquid phase) can be expressed as

$$\sigma^* = \sigma' + i\sigma'' = \sigma' + i\epsilon'\epsilon_0\omega \quad (18)$$

where σ' and σ'' represent the real and imaginary parts of the complex conductivity of the phase, respectively, and ϵ' is the related relative permittivity. In the numerical calculation, the fluid conductivity is $1 \times 10^{-4} \text{ S m}^{-1}$ and the relative permittivity ϵ' of water is 80; the conductivity and relative permittivity of solid phase are $5 \times 10^{-5} \text{ S m}^{-1}$ and 4, respectively. Thus, the complex conductivity of fluid phase σ_1^* is equal to $1 \times 10^{-4} \text{ S m}^{-1} + i 80\epsilon_0(2\pi f) \text{ S m}^{-1}$. The complex conductivity of the solid particle σ_2^* is $5 \times 10^{-5} \text{ S m}^{-1} + i 4\epsilon_0(2\pi f) \text{ S m}^{-1}$.

It was shown in previous research that the cell size (or model resolution) δ might influence the calculation (Zhan *et al.* 2010). Therefore, it is important to choose an appropriate δ value for the calculation. In this study, we determine δ by analysing its effect on the calculated porosity ϕ and surface-area-to-pore-volume ratio S_{por} . For a fixed particle radius of $35 \mu\text{m}$, five model resolutions ($\delta = 10, 5.3, 2.6, 1.3$, and $0.7 \mu\text{m}$) were used and the calculated ϕ and S_{por} are plotted in Figure 2. It is shown that ϕ is almost constant while δ varies between $10 \mu\text{m}$ and $0.7 \mu\text{m}$. The value of S_{por} continuously increases as δ decreases. However, when δ is smaller than $\sim 2 \mu\text{m}$, the change in the calculated S_{por} is insignificant. As a trade-off between accuracy and computation cost, we chose $\delta = \sim 1.4 \mu\text{m}$ for other calculations in this section.

For the effective conductivity of the sample with a simple cubic structure, Torquato (1985) has developed a theoretical model based on a perturbation expansion of the polarization. The model is based on a set of microstructure-dependent parameters, and truncation of the expansion yields an accurate approximation formula for the effective conductivity of the material (e.g., see

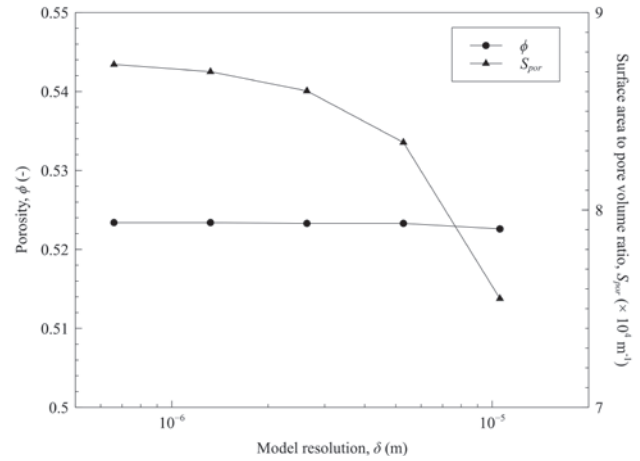


Figure 2 Calculated porosity and surface-area-to-pore-volume ratio of the sample in Figure 1 with different model resolutions.

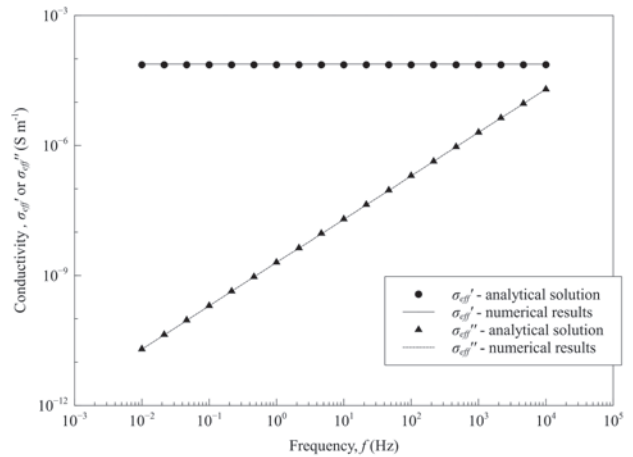


Figure 3 The complex effective conductivity of the sample in Figure 1 at different frequencies.

Cheng and Torquato 1997), which is expressed as (Torquato 1985)

$$\sigma_{eff}^* = \sigma_1^* \frac{1 + 2f_2\beta_1 - 2(1-f_2)\zeta_2\beta_1^2}{1 - f_2\beta_1 - 2(1-f_2)\zeta_2\beta_1^2}, \quad (19)$$

where f_2 is the volume fraction of the sphere. The three-point parameter ζ_2 is dependent on the microstructure of the material (e.g., porosity). For the simple cubic lattice structure in Figure 1, the ζ_2 values were reported in Torquato (1985) (see his Table 1). Parameter β_1 is related to the conductivity of the sample, and it has the following form:

$$\beta_1 = \frac{\sigma_2^* - \sigma_1^*}{\sigma_2^* + 2\sigma_1^*}, \quad (20)$$

where σ_1^* and σ_2^* are the complex conductivity values for phases 1 and 2, respectively. Apart from a trivial constant, parameter β_1 is equal to the dipole polarizability of a sphere with conductivity σ_2^* embedded in a matrix of conductivity σ_1^* [e.g., see equation (4) in Sihvola and Kong (1988)].

The effective complex conductivity of the sample in Figure 1 is calculated both numerically and analytically in the frequency range between 10^{-2} Hz and 10 kHz, and the results are shown in Figure 3. It appears that the complex conductivity calculated from numerical method is in good agreement with the analytical solution in this broad frequency range. To further test the numerical approach, we change the porosity of the sample ϕ by varying the radius but fixing the position of the particle. Same material properties were used, and the numerical results (at 1 Hz) are shown in Figure 4 where the porosity ϕ varies between 0.5 and 0.95. Equation (19) is also used to calculate the complex conductivity of the sample at $\phi = 0.9, 0.8, 0.7, 0.6$, and 0.5. The results are shown in Figure 4 with solid symbols. Again, the numerical solution matches the analytical solution very well, validating the applicability of the numerical approach.

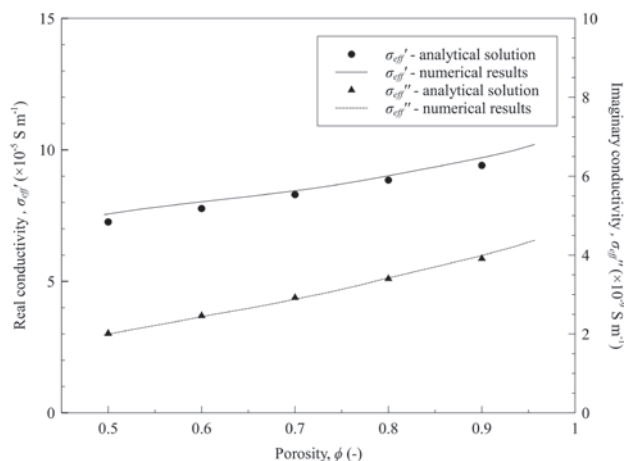


Figure 4 The complex effective conductivity (at 1 Hz) of samples having the simple cubic structure with different porosities.

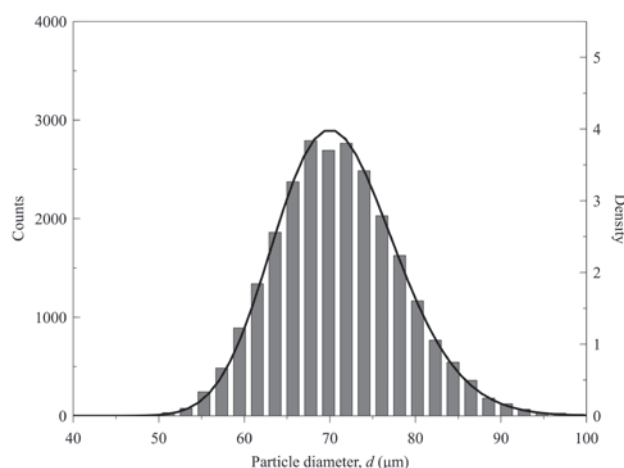


Figure 5 The particle size distribution of the glass beads sample. The solid curve is calculated from lognormal distribution with mean particle diameter $d = 70 \mu\text{m}$ and standard deviation $\bar{\sigma} = 0.1$. The histogram represents the particle size distribution of the numerical sample reconstructed in this study.

COMPARISON WITH EXPERIMENTS

In this section, we test whether the numerical approach can simulate the frequency-dependent IP response of geomaterials. We choose the glass beads sample reported in the work of Leroy *et al.* (2008) to compare with. The sample is numerically reconstructed, and its complex conductivity is then calculated using the aforementioned approach.

Glass beads sample

The sample in Experiment 1 in the work of Leroy *et al.* (2008) is used here. As reported, the particle size of the sample follows a lognormal distribution with mean particle diameter $d = 70 \mu\text{m}$ and the standard deviation $\bar{\sigma} = 0.1$ (Figure 5). The spherical glass beads were packed in a cylinder with 40% porosity and saturated with NaCl solution ($\sigma_w = 0.005 \text{ S m}^{-1}$). In the numerical simulation, we recreate the sample by randomly generating a number of spheres, the radii of which follow the same lognormal distribution (Figure 5). The spheres are packed in a cylindrical container by allowing them to fall in the gravitational field. The total number of the particles in the numerical sample is 16000 and the porosity is 0.39.

Numerical calculation

A cubic sample is taken from the packed cylindrical sample for the calculation of complex conductivity (see a 3D illustration in Figure 6). In the calculation, the fluid conductivity σ_w is assigned as 0.005 S m^{-1} , which is the same as the experimental value. The surface site densities associated with the diffuse and Stern layers are assumed identical, and their values were determined by fitting the calculated curve to experimental data as $10^{16} \text{ sites m}^{-2}$ (this corresponds to a surface charge density of $1.6 \times 10^{-3} \text{ C m}^{-2}$). It is noted that the value used here is in the same order with but slightly smaller than those used in the work of Leroy *et al.* (2008), in which 7.5×10^{16} and $6.5 \times 10^{16} \text{ sites m}^{-2}$ were used for the diffuse and Stern layers, respectively. The mobility values of Na^+ in the diffuse and Stern layers of the glass beads (silica) are assumed to be the same, i.e., $5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ (Revil 2013).

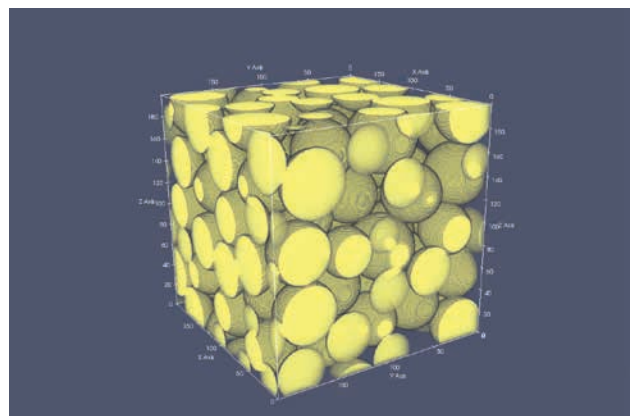


Figure 6 The 3D illustration of the reconstructed sample used in the calculation of the complex conductivity of the glass beads sample. The sample size is $375 \mu\text{m}$.

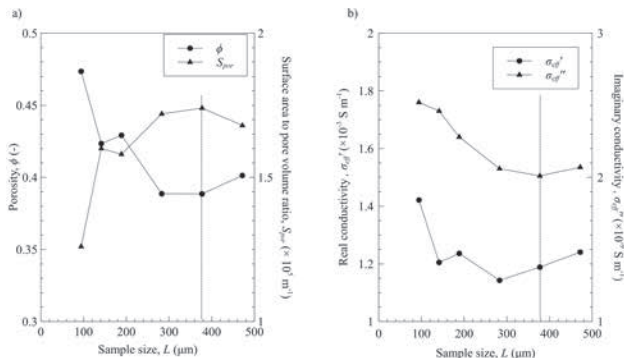


Figure 7 The influence of sample size on the calculated porosity, surface-area-to-pore-volume ratio, and complex conductivity of the reconstructed glass beads sample. The dash line indicates the sample size $L \approx 375 \mu\text{m}$, which was used in the subsequent calculations.

Coefficient M in equation (5) was estimated with equations (6), (7), and (8) by assuming room temperature (20°C); the estimated $M \approx 2$. The EDL-induced complex surface conduction C_s^* was upscaled using two ways: (1) converting C_s^* to σ_s^* (grain upscaling) and (2) converting C_s^* to $\Delta\sigma_w^*$ (pore upscaling). In either case, the associated phase will have an extra complex conductivity in addition to its intrinsic value as discussed before.

Previous research has shown that the sample size may influence the calculated physical/geometrical properties of geomaterials (Andr  et al. 2013b). In general, it is preferred to use a large sample. However, because of the computational limitation, a trade-off has to be made. In this study, we performed numerical calculations with different sample sizes and the trade-off was chosen according to the results (Figure 7). For all the calculations in Figure 7, a fixed model resolution was used, i.e., $\delta \approx 1.4 \mu\text{m}$. It is obvious that the calculated ϕ , S_{por} , σ_{eff}' , and σ_{eff}'' are influenced by the sample size L , as shown in Figure 7. However, when L is larger than $300 \mu\text{m}$, it seems that the variations in these calculations are quite small. As a balance between accuracy and computational cost, in the subsequent calculations (see the two subsequent sections), we chose $L = 375 \mu\text{m}$ and the cubic sample is discretised into 200^3 cells. This corresponds to a model resolution of $\sim 1.9 \mu\text{m}$.

Numerical results and comparison with experiments

The effective complex conductivity of the sample in Figure 6 is calculated in the frequency range between 10^{-1} Hz and 10 kHz, and the calculated phase shift $\theta = \text{atan}(\sigma_{\text{eff}}''/\sigma_{\text{eff}}')$ is plotted in Figure 8. The measured values of the glass beads sample are also plotted for comparison. In general, the numerical calculations well reproduce the trend of the measured phase spectra at both low (<100 Hz) and intermediate frequencies (>100 Hz). The characteristic frequency (e.g., peak frequency) of numerical curves is slightly smaller than that of the measured spectra. This is probably because the effect of diffuse layer polarization is underestimated in the calculation (i.e., M is underestimated). At frequencies larger than 100 Hz, the calculated phase is slightly

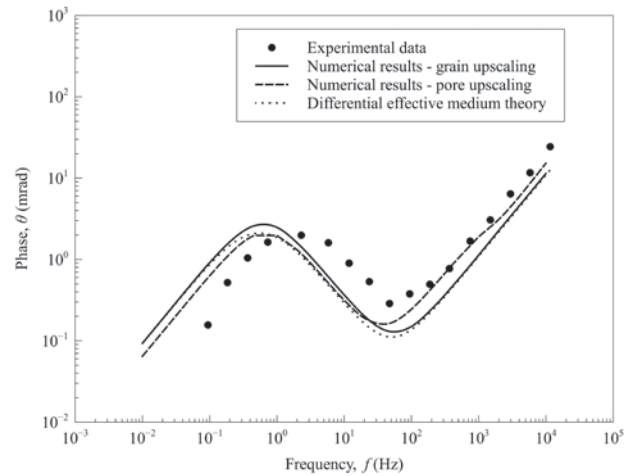


Figure 8 The measured and calculated phase of the glass beads sample. Experimental data are from Leroy et al. (2008). In the pore-scale calculation, the sample was numerically reconstructed with spheres with the same size distribution. Theoretical curve from the DEM is also plotted.

smaller than the experimental results. This might be because parasitic capacitance was involved in the experiment (Leroy et al. 2008). Grain upscaling and pore upscaling give similar results although noticeable difference can still be found, especially at low frequencies, as shown in Figure 8.

The conductivity model developed by Bussian (1983) is also used here for comparison with the proposed numerical approach. The model is based on the DEM theory, and its complex form is written as

$$\sigma_{\text{eff}}^* = \frac{\sigma_1^*}{F} \left(\frac{1 - \sigma_2^*/\sigma_1^*}{1 - \sigma_2^*/\sigma_{\text{eff}}^*} \right)^m, \quad (21)$$

where σ_1^* and σ_2^* are the complex conductivity values of the fluid and grains, respectively. The modelled phase with equation (21) is also plotted in Figure 8 as indicated by the dotted line. In the calculation, parameter m was 1.5 (Leroy et al. 2008) and the formation factor F was 3.9 (calculated as $F = \phi^m$). The complex fluid conductivity was identical to the value used in the numerical simulation. The complex conductivity of grains is the sum of σ_s^* [equation (10)] and the dielectric effect $i4\epsilon_s\epsilon_0\omega$. In calculating σ_s^* , the electrochemical properties of EDL are same as those used in the above numerical simulations and the characteristic particle size $\Pi = 35 \mu\text{m}$. In general, as shown in Figure 8, the phase spectra calculated using the DEM theory are quite similar to the numerical and experimental results, especially for results with grain upscaling.

EFFECT OF FLUID CONDUCTIVITIES

The fluid conductivity can influence the real conductivity of geomaterials. At high salinity, the surface conduction is insignificant; therefore, effective real conductivity σ_{eff}' is linearly proportional to the fluid conductivity (Rhoades, Raats and Prather 1976); the inverse of the slope is known as the intrinsic formation factor F .

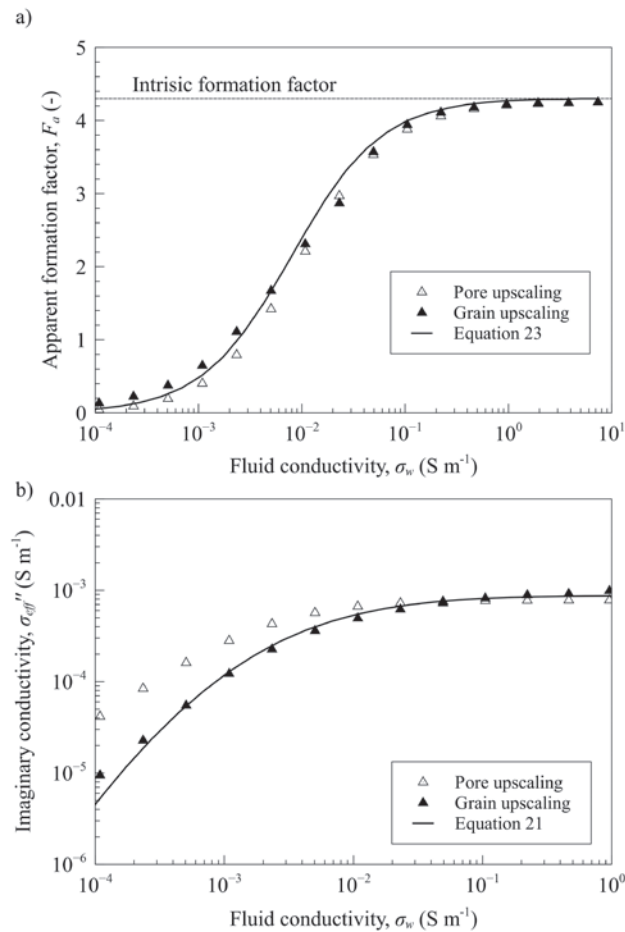


Figure 9 Complex conductivity of the granular sample in Figure 6 at different fluid conductivity values. The data points are from the numerical approach, and solid curves are from the DEM theory.

When the fluid conductivity σ_w decreases, the influence of surface conduction becomes noticeable. The $\sigma_{eff}' - \sigma_w$ relation becomes nonlinear, and the ratio σ_w / σ_{eff}' (i.e., apparent formation factor F_a) starts to deviate from the intrinsic formation factor F towards a smaller value. The salinity-dependent F_a has been observed in a range of geomaterials, including clear sandstone (Revil, Kessouri and Torres-Verdín 2014b), clayey sandstones (Niu, Revil and Saidian 2016b), and unconsolidated sediments (Weller, Slater and Nordsiek 2013). The fluid conductivity σ_w can also affect the low-frequency (<100 Hz) imaginary conductivity σ_{eff}'' of a geomaterial (Lesmes and Frye 2001). At low-to-intermediate salinities, σ_{eff}'' increases with σ_w ; after reaching a certain salinity, σ_{eff}'' keeps almost constant regardless of the increase in σ_w (Revil and Skold 2011; Weller and Slater 2012).

Numerical calculations

In this section, we use the developed numerical approach to calculate the complex conductivity of the granular material in Figure 6 and test whether it can reproduce the aforementioned trend. In the calculation, the salinity dependence of surface con-

ductance is considered. Instead of solving the equilibrium equations describing the complete chemical reactions on a silica surface in contact with NaCl solution (e.g., Leroy *et al.* 2008), we use a simplified model developed by Revil and Skold (2011) to account for the salinity effect on the surface conductance. The surface conductivity related to the Stern layer is expressed as

$$\Sigma_s^s = e\beta_{(+)}^s \Gamma_s^0 \frac{K_{Na} C_f}{10^{-pH} + K_{Na} C_f}, \quad (22)$$

where Γ_s^0 is the total surface site density and K_{Na} is the equilibrium constant associated with the adsorption of Na^+ at the surface of SiO^- sites on silicate grains. In the calculation, the following parameters were used: $\Gamma_s^0 = 5 \times 10^{18} \text{ m}^{-2}$, $K_{Na} = 10^{-3.25}$ (Revil and Skold 2011), and $pH = 7$. Both grain upscaling and pore upscaling were used. The calculated F_a and σ_{eff}'' are shown in Figures 9a and 9b, respectively. In Figure 9a, the intrinsic formation factor F is also plotted, which was numerically calculated by setting the surface conductance as zero.

The DEM-based equation (21) is also used here for comparison, and the results are shown in Figure 9. We defined the ratio of fluid conductivity σ_w and σ_{eff}' as the apparent formation factor F_a . At low frequency, equation (21) can be simplified by ignoring high-order terms expressed as (Niu *et al.* 2016a)

$$F_a = \frac{F\sigma_w}{a + \sigma_w}, \quad (23)$$

where a is equal to $m(1-1/F)\sigma_s$, with σ_s being the real part of σ_s^* . In the calculation, the following parameters were used: $a = 8 \times 10^{-3} \text{ S m}^{-1}$ and $F = 4.3$. Parameter a was determined by fitting equation (23) to the numerical results, as shown in Figure 9a.

Results and discussion

In Figure 9a, the apparent formation factor F_a calculated using the numerical method shows strong salinity dependence. At low salinities, F_a is lower than the intrinsic formation factor F , indicating a significant contribution of the surface conduction. When $\sigma_w > 5 \times 10^{-1} \text{ S m}^{-1}$, the calculated F_a is equal to F , implying the dominance of the bulk conduction. In general, the calculation from the developed numerical approach well reproduces the salinity-dependent real conductivity response of a geomaterial. Moreover, the numerical results fairly match the results from DEM theory, providing additional evidence for the validation of the numerical approach.

Figure 9b presents the imaginary conductivity of the sample. The numerical approach with either grain upscaling or pore upscaling results in salinity-dependent imaginary conductivity, which is in good agreement with trend observed in some published studies (e.g., Revil and Skold 2011; Weller and Slater 2012). The theoretical curve from the DEM theory [equation (21)] is also plotted in the figure, and it agrees well with the numerical results, especially for those calculated with grain upscaling. The consistence between published experimental trends and simulated results again supports the validation of the

proposed numerical approach. It is also noted that the critical fluid conductivity in Figure 9b ($\sim 0.01 \text{ S m}^{-1}$), at which the imaginary conductivity starts to decrease significantly, is roughly one order smaller than that observed in experiments (e.g., $\sim 0.1 \text{ S m}^{-1}$ in Revil and Skold 2011). The exact reason is unclear, and it might be due to the simplified treatment of the influence of fluid chemistry on surface conductance [i.e., equation (22)].

SUMMARY AND CONCLUSION

In this study, we present a pore-scale numerical approach for the modelling of the complex conductivity σ_{eff}^* of water-saturated granular materials. The effect of EDL on σ_{eff}^* is considered by assuming that the diffuse layer contributes to the electric conduction whereas the Stern layer is responsible for the electric polarization. The combined effect is quantified by the complex surface conductance, which is then converted to an apparent volumetric complex conductivity for solid particles (i.e., grain upscaling) or pore fluid (pore upscaling). The effective complex conductivity of the sample is determined by directly solving the FD representation of the Laplace equation in a representative elementary volume of the sample. Since the geometrical characteristic and electrical properties of each phase are involved in the simulation, the interfacial polarization can also be accounted for in the numerical modelling.

Numerical simulations are performed on a synthetic sample with a simple cubic structure and a glass beads sample. The calculated complex conductivity of the synthetic sample well matches the analytical solution, showing the accuracy of the developed approach. The pore-scale modelling also reproduces the key characteristics of the IP response of the glass beads sample, including the relaxations at low and high frequencies. These examples provide some evidence for the validation of the numerical simulation.

The developed numerical approach is used to calculate the complex conductivity response of a granular material at different salinities. It is shown that the proposed approach can simulate the salinity-dependent apparent formation factor, which increases with fluid conductivity towards the intrinsic formation factor at high salinities. The imaginary conductivity σ'' calculated from grain upscaling and pore upscaling at different salinities shows the similar trend as observed in published experimental studies. It is also in good agreement with the results from the DEM theory.

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